

4. ELECTRONIC CONFIGURATION AND PERIODIC TABLE

In many – electron atoms, not all electrons play the same role in terms of their physical and chemical properties.

4.1. Electronic configuration in the ground state

The **electronic configuration** describes the arrangement of electrons within AOs. The AOs are occupied according to three general rules that determine the electronic configuration of any atom.

4.1.1. Aufbau principle

In the ground state, electrons first occupy the AOs with the lowest energy level.

The energy level of the same subshells AOs are degenerate. Madelung's rule estimates the energy levels of each subshell.

4.1.2. Pauli exclusion principle

In 1925, the Austrian physicist Pauli (1945, NP in physics) postulated that 2 electrons in an atom cannot be described the same combination of four quantum numbers.

Subshell	s ($\ell = 0$)	p ($\ell = 1$)	d ($\ell = 2$)
Number of AOs			
Maximum number of electrons			

Table 4-1 Maximum number of electrons in each subshell

4.1.3. Hund's rule

In the same subshell, the electronic configuration with the greatest number of identical (parallel) m_s has the lowest energy.

4.1.4. Mapping

$n = 1$			
$n = 2$			

$n = 3$			

Table 4-2 Electronic configuration of atoms from H ($Z=1$) to Ar ($Z=18$)

4.1.5. Core and valence electrons

The electronic configuration of an atom reveals that electron have very different physical and chemical properties:

- Core electrons

Core electrons occupying the AOs of lowest energy levels have very strong interactions with the nucleus. The AOs occupied by these core electrons are located close to the nucleus. They therefore make no contribution to the atom's chemical properties. They usually appear in the electronic configuration of the previous rare gas.

- Valence electrons

Valence electrons occupying the AOs with higher energy levels have weaker interactions with the nucleus. The AOs occupied by these valence electrons are located at the edge of the electron shell. They therefore make a strong contribution to the atom's chemical properties. They usually complete the electronic configuration of the previous rare gas.

4.2. The periodic table of elements

The periodic table of elements is used to determine directly the valence electronic configuration of an atom.

4.2.1. Structure of the table

The periodic table of elements is organised into 18 columns, known as groups, and 7 rows, known as periods. The elements are arranged in ascending order of atomic number. This organisation reveals similar chemical properties amongst elements belonging to the same column.

The table is also divided into s, p, d, and f blocks, corresponding to the type of electron subshell being filled.

The first modern classification of the elements was proposed in 1869 by the Russian chemist Mendeleev (1906, almost NP of chemistry) : he arranged them mostly according to their atomic mass and noticed the periodic repetition of certain properties.

4.2.2. Valence electronic configuration

For atoms in the first 5 periods, the number of valence electrons corresponds to their position in the period. Atoms in the same column have the same number of valence electrons and similar valence electronic configurations.

Madelung's rule is strictly respected for the s and p blocks. However, exceptions are found for the d and f blocks.

1.008	Non-metal	H	Hydrogen	1
6.94	Metal	Li	Lithium	3
22.99	Metal	Na	Sodium	11
39.098	Metal	K	Potassium	19
85.468	Metal	Rb	Rubidium	37
132.905	Metal	Cs	Cesium	55
223	Metal	Fr	Francium	87
9.012	Metal	Be	Beryllium	4
24.305	Metal	Mg	Magnesium	12
40.078	Metal	Ca	Calcium	20
87.62	Metal	Sr	Strontium	38
137.327	Metal	Ba	Barium	56
226	Metal	Ra	Radium	88
44.956	Metal	Sc	Scandium	21
88.906	Metal	Y	Yttrium	39
47.867	Metal	Ti	Titanium	22
91.224	Metal	Zr	Zirconium	40
92.906	Metal	Hf	Hafnium	72
178.49	Metal	Rf	Rutherfordium	104
50.942	Metal	V	Vanadium	23
50.9415	Metal	Nb	Niobium	41
92.906	Metal	Ta	Tantalum	73
180.948	Metal	Db	Dubnium	105
51.996	Metal	Cr	Chromium	24
51.9961	Metal	Mo	Molybdenum	42
95.94	Metal	W	Tungsten	74
183.84	Metal	Sg	Seaborgium	106
54.938	Metal	Mn	Manganese	25
54.938045	Metal	Tc	Technetium	43
186.207	Metal	Re	Rhenium	75
186.207	Metal	Bh	Bohrium	107
58.933	Metal	Fe	Iron	26
57.935343	Metal	Ru	Ruthenium	44
101.07	Metal	Os	Osmium	76
190.23	Metal	Hs	Hassium	108
58.9332	Metal	Co	Cobalt	27
102.9055	Metal	Rh	Rhodium	45
192.227	Metal	Ir	Iridium	77
201.97	Metal	Mt	Meitnerium	109
63.546	Metal	Cu	Copper	29
107.8682	Metal	Ag	Silver	47
196.967	Metal	Au	Gold	79
200.59	Metal	Hg	Mercury	80
112.414	Metal	Cd	Cadmium	48
200.592	Metal	Pt	Platinum	78
208.98	Metal	Rg	Roentgenium	111
127.60	Metalloid	Te	Tellurium	52
127.60	Metalloid	Sb	Antimony	51
127.60	Metalloid	Bi	Bismuth	83
208.98	Metal	Po	Polonium	84
209	Metal	At	Astatine	85
210	Non-metal	Rn	Radon	86
15.999	Non-metal	O	Oxygen	8
15.9994	Non-metal	S	Sulfur	16
32.06	Non-metal	Se	Selenium	34
78.9718	Non-metal	Br	Bromine	35
78.9718	Non-metal	Kr	Krypton	36
35.45	Non-metal	Cl	Chlorine	17
35.45	Non-metal	P	Phosphorus	15
30.973762	Metalloid	As	Arsenic	33
114.818	Metal	In	Indium	49
114.818	Metal	Sn	Tin	50
207.2	Metal	Pb	Lead	82
208.98	Metal	Bi	Bismuth	83
209	Metal	Po	Polonium	84
210	Non-metal	At	Astatine	85
222	Non-metal	Rn	Radon	86
14.007	Non-metal	N	Nitrogen	7
14.007	Non-metal	P	Phosphorus	15
30.973762	Metalloid	As	Arsenic	33
127.60	Metalloid	Te	Tellurium	52
127.60	Metalloid	Sb	Antimony	51
127.60	Metalloid	Bi	Bismuth	83
208.98	Metal	Po	Polonium	84
209	Metal	At	Astatine	85
210	Non-metal	Rn	Radon	86
16.00	Non-metal	C	Carbon	6
12.011	Non-metal	C	Carbon	6
28.0855	Metalloid	Si	Silicon	14
72.630	Metalloid	Ge	Germanium	32
118.710	Metal	Sn	Tin	50
207.2	Metal	Pb	Lead	82
208.98	Metal	Bi	Bismuth	83
209	Metal	Po	Polonium	84
210	Non-metal	At	Astatine	85
222	Non-metal	Rn	Radon	86
1.00794	Non-metal	H	Hydrogen	1
6.941	Non-metal	Li	Lithium	3
22.98976928	Non-metal	Na	Sodium	11
39.0983	Non-metal	K	Potassium	19
85.4678	Non-metal	Rb	Rubidium	37
132.90545196	Non-metal	Cs	Cesium	55
223.0197897	Non-metal	Fr	Francium	87
9.012182	Non-metal	Be	Beryllium	4
24.30409	Non-metal	Mg	Magnesium	12
40.0784	Non-metal	Ca	Calcium	20
87.62	Non-metal	Sr	Strontium	38
137.327	Non-metal	Ba	Barium	56
226.0254	Non-metal	Ra	Radium	88
44.9559376	Non-metal	Sc	Scandium	21
88.90584	Non-metal	Y	Yttrium	39
47.8671	Non-metal	Ti	Titanium	22
91.2242	Non-metal	Zr	Zirconium	40
92.90638	Non-metal	Hf	Hafnium	72
178.49	Non-metal	Rf	Rutherfordium	104
50.94276	Non-metal	V	Vanadium	23
50.94386	Non-metal	Nb	Niobium	41
92.90638	Non-metal	Ta	Tantalum	73
180.94788	Non-metal	Db	Dubnium	105
51.99616	Non-metal	Cr	Chromium	24
51.99616	Non-metal	Mo	Molybdenum	42
95.94	Non-metal	W	Tungsten	74
183.84	Non-metal	Sg	Seaborgium	106
54.938045	Non-metal	Mn	Manganese	25
54.938045	Non-metal	Tc	Technetium	43
186.207	Non-metal	Re	Rhenium	75
186.207	Non-metal	Bh	Bohrium	107
58.933195	Non-metal	Fe	Iron	26
101.07	Non-metal	Ru	Ruthenium	44
101.07	Non-metal	Os	Osmium	76
190.23	Non-metal	Hs	Hassium	108
58.933195	Non-metal	Co	Cobalt	27
102.9055	Non-metal	Rh	Rhodium	45
192.227	Non-metal	Ir	Iridium	77
201.97	Non-metal	Mt	Meitnerium	109
63.546	Non-metal	Cu	Copper	29
107.8682	Non-metal	Ag	Silver	47
196.967	Non-metal	Au	Gold	79
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112.414	Non-metal	Cd	Cadmium	48
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208.98	Non-metal	Rg	Roentgenium	111
127.60	Non-metal	Te	Tellurium	52
127.60	Non-metal	Sb	Antimony	51
127.60	Non-metal	Bi	Bismuth	83
208.98	Non-metal	Po	Polonium	84
209	Non-metal	At	Astatine	85
210	Non-metal	Rn	Radon	86
15.999	Non-metal	O	Oxygen	8
15.9994	Non-metal	S	Sulfur	16
32.06	Non-metal	Se	Selenium	34
78.9718	Non-metal	Br	Bromine	35
78.9718	Non-metal	Kr	Krypton	36
35.45	Non-metal	Cl	Chlorine	17
35.45	Non-metal	P	Phosphorus	15
30.973762	Non-metal	As	Arsenic	33
114.818	Non-metal	In	Indium	49
114.818	Non-metal	Sn	Tin	50
207.2	Non-metal	Pb	Lead	82
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209	Non-metal	At	Astatine	85
210	Non-metal	Rn	Radon	86
16.00	Non-metal	C	Carbon	6
12.011	Non-metal	C	Carbon	6
28.0855	Non-metal	Si	Silicon	14
72.630	Non-metal	Ge	Germanium	32
118.710	Non-metal	Sn	Tin	50
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208.98	Non-metal	Bi	Bismuth	83
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132.90545196	Non-metal	Cs	Cesium	55
223.0197897	Non-metal	Fr	Francium	87
9.012182	Non-metal	Be	Beryllium	4
24.30409	Non-metal	Mg	Magnesium	12
40.0784	Non-metal	Ca	Calcium	20
87.62	Non-metal	Sr	Strontium	38
137.327	Non-metal	Ba	Barium	56
226.0254	Non-metal	Ra	Radium	88
44.9559376	Non-metal	Sc	Scandium	21
88.90584	Non-metal	Y	Yttrium	39
47.8671	Non-metal	Ti	Titanium	22
91.2242	Non-metal	Zr	Zirconium	40
92.90638	Non-metal	Hf	Hafnium	72
178.49	Non-metal	Rf	Rutherfordium	104
50.94276	Non-metal	V	Vanadium	23
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47.8671	Non-metal	Ti	Titanium	22
91.2242	Non-metal	Zr	Zirconium	40
92.90638	Non-metal	Hf	Hafnium	72
178.49	Non-metal	Rf	Rutherfordium	104
50.94276	Non-metal	V	Vanadium	23
50.94386	Non-metal	Nb	Niobium	41
92.90638	Non-metal	Ta	Tantal	

5. ATOMIC PROPERTIES AND PERIODICITY

5.1. Atomic mass

The **atomic mass** of an element is the total mass of a single atom, expressed in atomic mass units (amu, see 1.2.3). It is supposed to be equal to the total number of protons and neutrons in its nucleus, as the mass of electrons is negligible.

5.1.1. Measurement method

The atomic mass of an element can be determined by using a mass spectrometer.

5.1.2. Trend in the periodic table

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Most elements exist as a mixture of isotopes. Therefore, atomic mass is the weighted average of various isotope masses, based on their natural abundance.

Example for Chlorine:

Isotope	amu	natural abundance	
³⁵ Cl	34.97	75.77%	
³⁷ Cl	36.97	24.23%	
⁴⁰ Ar	39.96	99.60%	
³⁹ K	38.96	93.26%	
⁴¹ K	40.96	6.73%	

5.2. Atomic radius

The **atomic radius** of an element is defined as the distance from its nucleus to the outermost electron shell. It corresponds to the radius of the highest occupied AO. This distance can be determined using the x-coordinate of the maximum radial probability density of this AO (see 3.3.3). Beyond this distance, the radial probability density gradually decreases, making it difficult to determine this distance experimentally.

In contrast, structures composed of stacked identical atoms or ions in the solid state, as well as diatomic molecules, have regular geometric properties that allow their radius to be determined experimentally.

5.2.1. Measurement methods

Covalent radius is determined by measuring half the distance between the nuclei of 2 adjacent, identical atoms that are bonded together (e.g. H₂, O₂).

Metallic radius is determined by measuring half the distance between the nuclei of 2 adjacent, identical metal atoms in a metallic crystal (e.g. Cu, Au).

Van der Waals radius is determined by measuring half the distance between nuclei of 2 non-bonded adjacent, identical atoms in a solid crystal. The VdW radius is much larger than the covalent radius.

Ionic radius is determined by measuring the internuclear distance between 2 adjacent, identical anions or 2 adjacent anion and cation in an ionic crystal. The ionic radius of a cation (resp. an anion) is shorter (resp. longer) than the corresponding atomic radius.

5.2.2. Trend in the periodic table

H 37							He 140 ^{vdw}
Li 157	Be 112	B 88	C 77	N 74	O 66	F 64	Ne 154 ^{vdw}
Na 191	Mg 160	Al 143	Si 118	P 110	S 104	Cl 99	Ar 188 ^{vdw}
K 235	Ca 197	Ga 153	Ge 122	As 121	Se 117	Br 114	Kr 202 ^{vdw}
Rb 250	Sr 215	In 167	Sn 158	Sb 141	Te 137	I 133	Xe 216 ^{vdw}
Cs 272	Ba 224	Tl 171	Pb 175	Bi 170	Po 190	At -	Rn 220 ^{vdw}

Table 5-1 Atomic radii (in pm)

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5.3. Ionisation energy

The **ionisation energy** of an element is defined as the minimum quantum of energy required to remove the outermost electron from an atom in the gas phase. The value is expressed in kJ.mol⁻¹ in chemistry.

5.3.1. Measurement methods

Ionisation energy can be determined by atomic spectroscopy (see 2.5) or by PhotoElectron Spectroscopy (PES).

5.3.2. Trend in the periodic table

H							He 2372
Li 520	Be 900	B 801	C 1086	N 1402	O 1314	F 1681	Ne 2081
Na 496	Mg 738	Al 578	Si 787	P 1012	S 1000	Cl 1251	Ar 1521
K 419	Ca 590	Ga 579	Ge 762	As 944	Se 941	Br 1140	Kr 1351
Rb 403	Sr 549	In 558	Sn 709	Sb 831	Te 869	I 1008	Xe 1170
Cs 376	Ba 503	Tl 589	Pb 716	Bi 703	Po 812	At -	Rn 1037

Table 5-2 Ionisation energies (in kJ.mol⁻¹)

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5.4. Polarisability

A **polar entity** has an asymmetric distribution of electron density: the barycentres of the negative and the positive charges are not co-located. This entity can be modelled as two equal and opposite charges q separated by a distance d . This simplification allows us to describe the polar entity using classical electrostatics:

$$\vec{\mu} = q\vec{d} \text{ (in D)}$$

The **polarisability** of an element is defined as how easily the electron shell of an atom can be distorted by an external electric field to form a polar entity.

5.4.1. Measurement methods

Polarisability can be determined by measuring how a substance's refractive index varies when placed in an external electric field.

5.4.2. Trend in the periodic table

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5.5. Electronegativity

The number of valence electrons (see 4.1.5) alone is unable to describe all the chemical properties of atoms. When two atoms are bonded together, the distribution of electron density between them is not always symmetric. Some atoms attract electrons more strongly than others.

A—A	A—B

Electronegativity χ is defined as an atom's ability to attract electron density towards itself when bonded to another atom. It is a relative quantity without unit.

5.5.1. Measurement methods

There are several definitions of electronegativity:

- Pauling definition

The Pauling definition is based on a comparison of bond dissociation energies of the diatomic molecules A—A, B—B and A—B. It is an experimental quantity.

Diatomic molecule	H—H	F—F	H—F
Bond dissociation energy (in kJ.mol ⁻¹)	436	158	567
Diatomic molecule	H—H	I—I	H—I
Bond dissociation energy (in kJ.mol ⁻¹)	436	151	299

Table 5-3 Bond dissociation energies of diatomic molecules

- Allred-Rochow definition

The Allred-Rochow definition is derived directly from the strength of the interaction between the nucleus and the outermost electron. It is a theoretical quantity.

5.5.2. Trend in the periodic table

H 2.2							He -
Li 0.98	Be 1.57	B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne -
Na 0.93	Mg 1.31	Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar -
K 0.82	Ca 1.0	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr -
Rb 0.82	Sr 0.95	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe -
Cs 0.79	Ba 0.89	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn -

Table 5-4 Pauling electronegativities

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